AN INJECTIVE HULL OF THE RESIDUE FIELD SATISFIES THE DESCENDING CHAIN CONDITION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The identification of the endomorphism ring of the injective hull with the completion of the local ring. Invoked inside the counted proof (source line 1113). Makes each hom module in the chain a module over the Noetherian completion.
- The Noetherianity of the completion of a Noetherian local ring. Invoked inside the counted proof (source lines 1115-1116). Stops the ascending chain of hom modules.
- The description of the hull as the union of the submodules killed by the powers of the maximal ideal. Invoked inside the counted proof (source lines 1119-1121). Provides a nonzero element annihilated by the maximal ideal in any nonzero quotient.

Source reference. Original reference algebra-lemma-completion-Noetherian-Noetherian: sources/stacks-dualizing/pinned-original-source.tex, source lines 1116–1116. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-completion-Noetherian-Noetherian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-endos: sources/stacks-dualizing/pinned-original-source.tex, source lines 1083–1083. Navigation pointer only: the counted text refers to the Stacks item lemma-endos, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-union-artinian: sources/stacks-dualizing/pinned-original-source.tex, source lines 1042–1042. Navigation pointer only: the counted text refers to the Stacks item lemma-union-artinian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $(R, \mathfrak{m}, \kappa)$ be a Noetherian local ring. Let E be an injective hull of κ over R. Then E satisfies the descending chain condition.*

Proof. If $E \supset M_1 \supset M_2 \supset \ldots$ is a sequence of submodules, then

$$\mathrm{Hom}_R(E, E) \to \mathrm{Hom}_R(M_1, E) \to \mathrm{Hom}_R(M_2, E) \to \ldots$$

is a sequence of surjections. By Lemma lemma-endos each of these is a module over the completion $R^\wedge = \mathrm{Hom}_R(E, E)$. Since $R^\wedge$ is Noetherian (Algebra, Lemma algebra-lemma-completion-Noetherian-Noetherian) the sequence stabilizes: $\mathrm{Hom}_R(M_n, E) = \mathrm{Hom}_R(M_{n+1}, E) = \ldots$. Since E is injective, this can only happen if $\mathrm{Hom}_R(M_n/M_{n+1}, E)$ is zero. However, if M_n/M_{n+1} is nonzero, then it contains a nonzero element annihilated by $\mathfrak{m}$, because E is $\mathfrak{m}$-power torsion by Lemma lemma-union-artinian. In this case M_n/M_{n+1} has a nonzero map into E, contradicting the assumed vanishing. This finishes the proof. $\square$